Statistic 403/603

Mid-term exam 2

NAME:

STUDENT NUMBER:

SIGNATURE:

INSTRUCTIONS:

1. Do not panic. It rarely helps.
2. Read all questions carefully.
3. Calculators are permitted (none that have communication ability).
4. Where computations are required, you should show work.
5. If a question calls for an explanation, provide one that is complete.
6. Cross out any material you do not wish to have considered.
7. **Correct answers with insufficient justification or accompanied by additional incorrect statements will not receive full credit.**
8. Correct/incorrect answers without clearly displayed work cannot be considered for partial credit.
9. You may lose marks if your explanations are incomplete, missing or poorly worded.

In accordance with the SFU Academic Honesty Policy (T 10.02), academic dishonesty in any form will not be tolerated. Prohibited acts include, but are not limited to, the following:

- making use of any books, papers or electronic devices that allow communication,
- speaking or communicating with other students who are writing exams,
- copying the work of other students or purposely exposing written papers to the view of other students

Formulas:

Simple random sampling (SRS)

Sample mean: $\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$

Sample standard deviation: $s = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}}$

Standard error of the mean: $se(\bar{y}) = \sqrt{(1 - \frac{n}{N})\frac{S^2}{n}}$

Estimate of the population total: $\hat{\tau} = N\bar{y}$

Standard error for the estimate of the population total: $se(\hat{\tau}) = N\sqrt{(1 - \frac{n}{N})\frac{S^2}{n}}$

Standard error for the estimate of the population proportion: $se(\hat{p}) = \sqrt{(1 - \frac{n}{N})\frac{P(1-P)}{n-1}}$

Sample size for SRS: $n \geq \frac{z_{\alpha/2}^2 s^2}{e^2 + \frac{z_{\alpha/2}^2 s^2}{N}}$

Stratified random sampling:

Estimate of the population mean (stratified random sampling): $\hat{\mu}_{str} = \bar{y}_{str} = \sum_{h=1}^{H} \frac{N_h}{N}\bar{y}_h = \sum_{h=1}^{H} W_h \bar{y}_h$

Standard error of stratified random sampling estimate of the pop. mean: $se(\mu_{str}) = \sqrt{\sum_{h=1}^{H} W_h^2 \frac{S_h^2}{n_h}(1 - n_h/N_h)}$

Cluster sampling:

Cluster sampling estimate of population mean: $\hat{\mu} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}$

Standard error of the cluster sampling estimator of the pop. mean: $se(\hat{\mu}) = \sqrt{\frac{1}{\bar{M}^2}\frac{S_{dif}^2}{n}(1 - n/N)}$

Cluster sampling estimator of pop. total: $\hat{\tau} = \hat{\mu}M$

Cluster sampling standard error for the estimator of the population total: $se(\hat{\tau}) = se(\hat{\mu})M$

1. (5 marks) The Betty Crooker Baking Company has two methods (call these methods A and B) for making chocolate cupcakes (a very nice desert). A baker decides to run an experiment to compare the fluffiness of the cupcakes resulting from each of the two methods (fluffiness will be measured by the height of a cupcake).

   To run the experiment, the baker randomly selects a method and uses the ingredients to make the cupcake batter. The batter is poured into cupcake tray and 12 cupcakes are baked. Next the other method is selected, and the ingredients are used to make the cupcake batter for the second method. The batter is poured into cupcake tray and 12 cupcakes are baked. At the end of baking, the heights of all 24 cupcakes is measured (12 cupcakes for each method).

   a. What is the factor(s) for this experiment?

   Method

   b. What are the levels of the factor?

   Levels: A and B

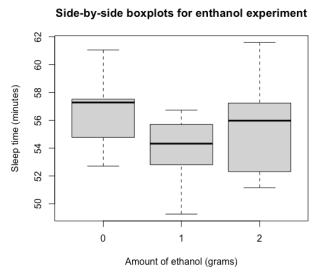
   c. What is the observation unit?

   Cupcake

   d. The is a major problem with this experiment. What is it and why is it a problem?

   There are a few answers, but the answer must be related to pseudo-replication. For example, the experiment does not have replicates of each treatment since all 12 cupcakes of each type came from the same batch of batter. If one of the two batches is mixed poorly, all 12 cupcakes might be affected.

2. (20 marks) Many people report sleep difficulties after a night of social drinking. To investigate this problem, an experiment was run where ethanol was administered to rats. A sample of 27 rats was selected and each rat was given a dose of ethanol (Doses: 0 g ethanol, 1 g ethanol, or 2 g ethanol). This was a balanced design. The sleep time (minutes) for each rat was then recorded.



a. Looking at the side-by-side boxplots, would you expects to observe a significant difference in the mean sleep time for each group (explain why or why not)?

Looking at the box-plots, we see that the intervals for the middle 50% of observations for each treatment overlap.

The medians for each group do not differ by much.

I would not expect the treatment means to be significantly different.

b. An ANOVA model will be fit to the data. Write out the ANOVA model and state the corresponding assumptions.

$$y_{ij} = \mu_i + \epsilon_{ij};\ \ i = 1,2,3;\ \ j = 1, 2, \dots 9$$

Where y_{ij} is the observed sleep time for the jth rat receiving treatment i
The ϵ_{ij}'s are independent and identically distributed where $\epsilon_{ij} \sim N(0, \sigma)$; $i = 1,2,3;\ \ j = 1, 2, \dots 9$

c. The model was fit in R and the following was recorded:

```
> mymodel=aov(sleep.time~ethanol)
> anova(mymodel)
Analysis of Variance Table

Response: sleep.time
          Df Sum Sq Mean Sq F value Pr(>F)
ethanol    2  31.72 15.8599  2.0183 0.1548
Residuals 24 188.59  7.8579
```

State the null and alternate hypotheses that the ANOVA model is meant to assess.

$H_0: \mu_1 = \mu_2 = \mu_3\ vs\ \ H_1: At\ least\ one\ fo\ the\ \mu_i' sare\ different$ where μ_i is the mean sleep time of rats for the ith treatment.

d. The p-value here is 0.1548. What is the meaning of a p-value? Explain in the context of this question.

The p-value is the probability of observing a test statistic as extreme or more extreme when the null hypothesis is true.

In this context it means that if the null hypothesis were true (the mean sleep time under all three treatments is the same), then the probability of observing an F-statistic as large or larger than 2.0183 is 0.1548.

e. What do you conclude (why) for this experiment?

If α=0.05, the since the p-value > α, we do not reject the null hypothesis. Therefore, we conclude that the data do not give evidence that the mean sleep time for rats is different for the doses of 0 g ethanol, 1 g ethanol, or 2 g ethanol.

f. Based on part e., what type of error could have been committed (explain)?

Since we did not reject the null hypothesis, we could have failed to do so when, in fact, the null is not true. This is a type II error.

g. The experimenter did a calculation after the fact and estimated that the power to detect a significant effect of 2 minutes for this experiment design was 0.40 (sometimes called 40% power). Explain the following to a statistical beginner in the context of this question: What is the meaning of 40% power?

If this experiment was repeated many times, conducted and analyzed in the same way, and the difference between the largest and smallest treatment means was 2 minutes, the probability of correctly rejecting the null hypothesis is 0.40 or 40% of the time.

h. Suppose the experiment above was used as a pilot study. What would the likely recommendations regarding sample sizes for each treatment have been (why?)? You do not have to report a number.

Correctly rejecting the null hypothesis 40% of the time is not very good. The experimenter would likely want this to be higher. To increase the power, they would have to take larger sample sizes for each group to increase the power.

i. Suppose the experimenter wanted to construct a 95% confidence interval for the difference in the mean sleep time under the 0 grams and 2 grams of ethanol treatments. **Estimate** the width of the resulting confidence interval.

The CI would be $\bar{y}_2 - \bar{y}_0 \pm t_{0.975}(16) s_p \sqrt{\frac{1}{n_2} + \frac{1}{n_0}}$ where $t_{0.975}(16)$ is the 97.5th percentile of the T-distribution with 16 degrees of freedom. We can replace $t_{0.975}(16)$ with the corresponding normal percentile of 1.96 (I will use 2, but either is fine for this).
s_p is the pooled standard deviation that estimates the population standard deviation, σ. The square root of the mean square error is the pooled standard deviation over the three treatments, and we can use this in place of s_p

So, the margin of error is $t_{0.975}(16) s_p \sqrt{\frac{1}{n_2} + \frac{1}{n_0}} \approx 2\sqrt{7.85}\sqrt{\frac{1}{9} + \frac{1}{9}} = 2.64$

The width of the CI would be about 2(2.64) = 5.28 minutes

3. (8 marks) Inland fish farming is thought by some to be a more sustainable way to have salmon in our food supply without harming wild salmon stocks. The weights of salmon using 2 varieties of fish food (F1 and F2) are to be compared using a study at a BC fish farm. Within the farm, there are 16 large, separated salmon pens containing baby, sockeye salmon arranged in a 4 x 4 grid pattern at the fish farm. Eight replications of each treatment are to be done. The treatments are to be assigned to the pens where the salmon grow. After growth, the average weight of salmon in a pen is measured (assume the number of salmon per pen is the same).

a. Show an experimental layout that uses a completely randomized design to investigate the effect of the factor. What are the main advantages of this design?

F2	**F1**	**F1**	**F1**
F2	**F2**	**F1**	**F2**
F1	**F1**	**F2**	**F2**
F2	**F2**	**F1**	**F1**

The advantage is that this is the easiest to conduct and requires the least complicated analysis.

b. There is a busy road that runs the south side of the 4 x 4 grid of pens. The scientists feel that there may be an impact on the yield of being close to the road. Show an experimental layout that uses a randomized block design to investigate the effect of the factor. Give enough details so that a technician can follow your plan (what are the blocks, treatments, experimental units, and how are treatments randomized to experimental units). What are the main advantages of this design in comparison to the design in part a?

Block 1	**F1**	**F1**	**F2**	**F2**
Block 2	**F2**	**F2**	**F1**	**F2**
Block 3	**F1**	**F2**	**F1**	**F2**
Block4	**F2**	**F1**	**F1**	**F2**

ROAD

The pens are arranged in a 4 x 4 grid.
The blocks are the 4 rows of pens parallel to the road
There are two treatments (F1 and F2) that are to be randomized to experimental units (pens).

Since there are two treatments and 4 pens per block, each treatment will be applied to two pens per block. To do so, I will place 4 balls in a bag, 2 labeled F1 and 2 labeled F2. I will select a ball from the bag, the label on the first ball taken will be given to the first pen. The procedure is repeated until all 4 balls have been selected and assigned to pens 1-4.

Main advantage is that the impact of the road is mitigated by the block. Each treatment is run at similar distances from the road. This will increase the power if the road has an (additive) impact on the response.